Range of α-particles & setting up $\alpha - \gamma$ coincidence

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I. OBJECTIVE

1. To determine the range of α-particles using a ^{241}Am source.
2. To set up and study the $\alpha - \gamma$ coincidence spectra for a ^{241}Am source.

II. INTRODUCTION

A heavy charged particle, such as an α-particle loses energy in matter primarily by excitation and ionization of atoms along its path. The energy loss occurs through statistical interactions with atoms of the medium. The average distance travelled by a particle before it loses all of its energy is called *range* of the particle [1].

In the first part of the experiment, we determine the range of α-particless. Instead of varying the distance of the source and detector, we fixed the distance between source and detector and varied the pressure in the chamber, which in turn varied the effective distance in air at STP. We study the variation of energy of the α particle with effective distance and extrapolate the linear fit to effective distance corresponding to zero energy to determine the range [2].

In the second part, we will study the $\alpha - \gamma$ coincidence and the correlation heatmaps between the two spectra. The coincidence technique can provide additional information about the decay scheme of ^{241}Am.

III. PROGRESS

A. Week 1

- We studied the reading materials related to the experiment, including the energy loss of charged particles and particle range.
- We familiarized ourselves with the instruments used in the experiment and their functions and limitations.
- We installed and set up the open-source Cspark MCA software for data acquisition.

B. Week 2

- We attempted to use the amplifier component of the PHYWE pulse height analyser, but it did not function.
- We then used the PHYWE MCA as an amplifier and observed the amplifed waveforms on the oscilloscope. We found that the signal amplitude was too small for acquisition with the Cspark MCA
- We found that the gain of the PHYWE MCA could be adjusted digitally. We therefore installed the PHYWE Measure software, set the gain to its maximum value, and obtained a usable signal.
- Subsequently, we use the Cspark MCA for data acquisition

C. Week 3

- We collected data with the source and detector at a distance of 0.3 cm.
- We found that the effective distances corresponding to our pressure range of approximately $1-500$ $mbar$ are too small. The residual energy - distance dependence was found to be non-linear.
- Therefore, we increased the distance between source and detector to 3 cm and collected data over the same range of pressure.
- We calculated the corresponding effective distances and plotted alpha residual energy vs effective distance in air at STP (see Figure 1). We calculated the x-intercept of the linear fit to determined the range of α-particles as $R_\alpha = 2.61 \pm 0.01$ cm.

IV. FUTURE PLAN

We are currently learning to use the FPGA-based data acquisition system and working on the gamma detector setup. In the next few weeks, we plan to set up and test the apparatus for $\alpha - \gamma$ coincidence. We will then acquire coincidence data and analyse α and γ spectra. We aim to study the correlation heatmaps between the α and γ spectra and verify the decay scheme of ^{241}Am.

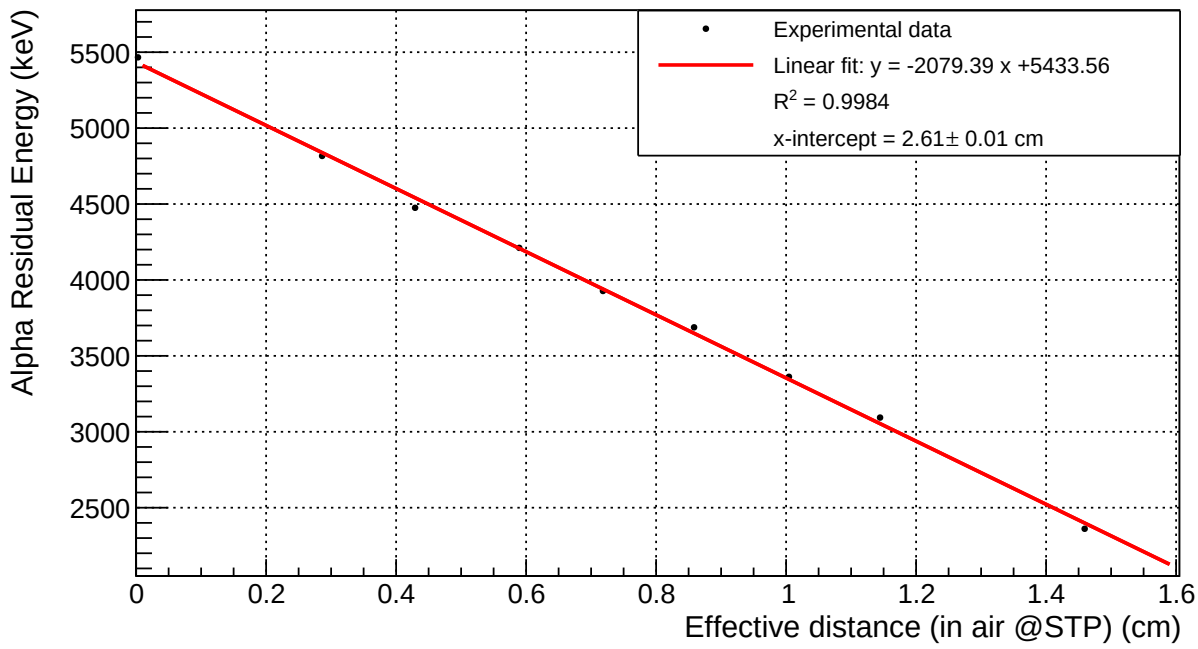


FIG. 1. Measured residual energy of α-particles as a function of effective distance in air at STP. The solid line represents the linear fit to the experimental data, whose x-intercept gives the estimated range of the α-particles.

[1] William R. Leo. *Techniques for Nuclear and Particle Physics Experiments: A How-to Approach*. Springer-Verlag, Berlin, Heidelberg, 2 edition, 1994.

[2] Andrius Poškus. Energy loss of alpha particles in gases. Technical Report Experiment No. 2, Vilnius University, Faculty of Physics, Department of Solid State Electronics, Laboratory of Applied Nuclear Physics, February 2010.

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